

Functional Annotation Report: *dapF* (Diaminopimelate epimerase, DapF) in *Pseudomonas putida* KT2440

Target: gene `dapF` / locus `PP_3790` / UniProt **Q88GD4** (`dapF_Q88GD4`) **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950 / KT2440), PSEPK **Enzyme:** Diaminopimelate epimerase (DAP epimerase), EC 5.1.1.7; "PLP-independent amino acid racemase" **Family:** Diaminopimelate epimerase family — Pfam **PF01678** (DAP_epimerase); InterPro **IPR001653**; HAMAP-Rule **MF_00197**; TIGRfam **TIGR00652** (dapF)

1. Summary (Answer to the Research Question)

dapF (Q88GD4, PP_3790) encodes **diaminopimelate epimerase (DapF)**, a soluble **cytoplasmic** enzyme that catalyzes the reversible **stereoinversion of LL-2,6-diaminopimelate (LL-DAP) to meso-2,6-diaminopimelate (meso-DAP)** — the penultimate reaction of the diaminopimelate (DAP) branch of L-lysine biosynthesis (EC 5.1.1.7). DapF is a **cofactor-independent (pyridoxal-5'-phosphate-independent) epimerase** that uses an unusual **two-base "1,1-proton-transfer" mechanism** rather than a PLP cofactor. In the classical enzymes this two-base pair is two active-site cysteines (thiol/thiolate); however, **a key finding of this report is that Q88GD4 appears to be a member of the recently described DapF-SS subclass** — its two catalytic-position residues are **serines (Ser70 and Ser205), not cysteines** — whereas its *P. putida* paralog Q88CF3 (PP_5228) is a canonical DapF-CC (Cys75/Cys219). Its product, *meso*-DAP, has two essential fates: (i) direct incorporation into the peptidoglycan stem peptide (position 3) of the Gram-negative cell wall, and (ii) decarboxylation by LysA (DAP decarboxylase) to yield L-lysine for protein synthesis. In *P. putida* KT2440, the genome encodes the complete **succinylase branch** that produces DapF's LL-DAP substrate and **lacks the *ddh* (meso-DAP dehydrogenase) bypass**, so DapF activity is the **obligate** route to *meso*-DAP. The functional assignment for Q88GD4 is by strong homology (UniProt evidence level PE:3) to biochemically and structurally characterized orthologs; the mechanistic, structural, and kinetic evidence below derives from those orthologs (*Escherichia coli*, *Haemophilus influenzae*) and from the DapF-SS characterization of Lamer et al. 2025.

2. Gene/Protein Identity Verification

The gene symbol **dapF**, the enzyme name **diaminopimelate epimerase**, EC 5.1.1.7, the "PLP-independent amino acid racemase" alternate name, the **DAP_epimerase family (PF01678/IPR001653)**, and the organism (*P. putida* KT2440) are **fully mutually consistent** across UniProt and the primary literature. Retrieval of the Q88GD4 sequence and annotations confirmed:

- A **268-residue** protein, gene `dapF`, locus `PP_3790`.
- UniProt keywords: **Cytoplasm, Lysine biosynthesis, Amino-acid biosynthesis, Isomerase**.
- UniProt FUNCTION: "*Catalyzes the stereoinversion of LL-2,6-diaminopimelate (L,L-DAP) to meso-diaminopimelate (meso-DAP), a precursor of L-lysine and an essential component of the bacterial peptidoglycan.*"
- UniProt PATHWAY: "*L-lysine biosynthesis via DAP pathway; DL-2,6-diaminopimelate from LL-2,6-diaminopimelate: step 1/1.*"
- The two conserved pK-modulating residues (His147, Glu196) that flank the catalytic center are present; however, UniProt assigns **no "Active site" cysteine residues** to Q88GD4 (unlike the *E. coli* enzyme and the Q88CF3 paralog), an early hint that the catalytic residues are non-cysteine (see §4).

Conclusion: the target is unambiguously a diaminopimelate epimerase. No conflicting/paralogous-symbol literature issue was encountered. (Note: the *phzF* phenazine-biosynthesis protein shares the DapF fold but is a distinct isomerase lacking the catalytic cysteines —

P 15545603

— and is *not* the target here.)

3. Primary Function: The Catalyzed Reaction and Substrate Specificity

Reaction: LL-2,6-diaminopimelate $\rightleftharpoons$ meso-2,6-diaminopimelate (a 1,1-proton-transfer stereoinversion at one of the two α -stereocenters of the symmetric diamino-diacid DAP). This is the **penultimate step of L-lysine biosynthesis**: "D,L-Diaminopimelate is generated from the corresponding L,L-isomer by the dapF-encoded epimerase" (Koo & Blanchard 1999,

P 10194362

"Diaminopimelate (DAP) epimerase catalyzes the stereoinversion of ll-DAP to meso-DAP, a precursor of l-lysine and an essential component of the bacterial peptidoglycan" (Pillai et al. 2007,

P 17889830

Substrate specificity (experimental, orthologs): The enzyme is highly specific for the DAP scaffold. Purified *E. coli* LL-DAP epimerase epimerizes LL-DAP with **$K_m \approx 0.26$ mM**; substrate-analog studies show **N-hydroxy-DAP** is the most potent competitive inhibitor (**$K_i = 0.0056$ mM**), lanthionine isomers inhibit with $K_i = 0.18\text{--}0.42$ mM, N-amino-DAP is weak ($K_i = 2.9$ mM), and 4-methylene-DAP is noncompetitive ($K_i' = 0.95$ mM) (Lam et al. 1988,

P 3042781

α -Methyl amino acids act as slow-binding inhibitors (Lamer et al. 2025,

P 40299312

This SAR indicates a tightly complementary active site that recognizes the α -amino/ α -carboxylate groups flanking the epimerized center and discriminates strongly against non-DAP amino acids. (Some family members show minor promiscuous activity on lysine/related amino acids, but DAP is the physiological substrate.)

4. Catalytic Mechanism and Structure

Cofactor-independent two-cysteine mechanism. Unlike alanine or glutamate racemases, DapF requires **no PLP or metal cofactor**. Catalysis relies on **two active-site cysteines**, one acting as a base (thiolate) to abstract the C α proton and the other as an acid (thiol) to re-donate a proton to the opposite face, inverting stereochemistry: - Cirilli et al. 1998

([P 9843410](#)):

in *H. influenzae* DapF, "Cys73 of the amino terminal domain is found in disulfide linkage, at the domain interface, with Cys217 of the carboxy terminal domain, and we suggest that these two cysteine residues in the reduced, active enzyme function as the acid and base in the mechanism." - Pillai et al. 2007

([P 17889830](#)):

DapF "belongs to the group of PLP-independent amino acid racemases that function through a rather unusual mechanism involving two cysteines acting in concert as a base (thiolate) and an acid (thiol)." - Koo & Blanchard 1999

([P 10194362](#)):

pH-dependence and kinetic isotope effects support two catalytic groups ($pK \approx 6-8.5$) and a two-base mechanism.

Recent work has even engineered/discovered natural variants that use paired **serines** (DapF-SS) in place of cysteines with similar kinetics, underscoring the generality of the two-base chemistry (Lamer et al. 2025,

[P 40774471](#)).

Architecture and conformational cycle. DapF is a **monomeric, two-domain** protein with internal pseudo-2-fold symmetry: "The 274 amino acid enzyme consists of two structurally homologous domains, each containing eight beta-strands and two alpha-helices" (Cirilli et al. 1998,

[P 9843410](#)),

placing one catalytic cysteine from each domain on opposite faces of the substrate Ca. Catalysis involves a **substrate-induced domain closure**: "Substrate binding to the C-terminal domain triggers the closure of the N-terminal domain coupled with tight encapsulation of the ligand... and expulsion of water molecules with concomitant desolvation of the thiolate base. This structural rearrangement is critical for catalysis" (Pillai et al. 2007,

[P 17889830](#)).

Solvent exclusion elevates the effective basicity of the thiolate and enforces strict stereocontrol, preventing non-productive racemization.

Mapping to Q88GD4 — a DapF-SS (serine-serine) variant. A precise residue-level mapping to the *experimentally* annotated *E. coli* DapF (UniProt P0A6K1, PE:1) was performed. In DapF the catalytic residue lies **immediately N-terminal to an invariant substrate-binding motif**: *E. coli* has catalytic **Cys73**–Gly74–Asn75 ("GN" binding site) and **Cys217**–Gly218–Ser219 ("GS" binding site). The *P. putida* paralog **Q88CF3 (PP_5228)** is annotated identically as a canonical **DapF-CC** with catalytic **Cys75/Cys219** (CGN and CG-T motifs). In the **target Q88GD4 (PP_3790)**, the equivalent catalytic positions — unambiguously fixed by its conserved binding motifs **GN (residues 71–72)** and **GT (residues 206–207)** — are **Ser70 and Ser205, not cysteines**. The pK-modulating His147 and Glu196 are retained, and UniProt assigns **no catalytic cysteine** to Q88GD4 (whereas it does for both *E. coli* and the paralog). The cysteines present in Q88GD4 (positions 19, 24, 33, 78, 103, 148, 210, 252, 263) are offset from the catalytic register (e.g., Cys78/Cys210 lie +8/+5 residues downstream of the true catalytic positions) and are **not** the catalytic residues.

This signature — full DAP-epimerase fold and substrate-binding residues, but paired **serines** at the two catalytic positions — matches the recently characterized **DapF-SS** subclass: Lamer et al. 2025

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P 40774471

) described "enzymes of the cofactor-independent racemase and epimerase class that use paired catalytic serines (DapF-SS) instead of cysteines," which "catalyze reversible epimerization of DAP with similar kinetic parameters to that of DapF-CC enzymes" and were found across "thousands of other predicted DapF-SS enzymes from diverse bacterial phyla." Q88GD4 is therefore best classified as a **putative DapF-SS diaminopimelate epimerase**, expected to perform the same LL-DAP $\rightleftharpoons$ meso-DAP reaction via an analogous two-base mechanism using serine hydroxyls in place of thiols. (This is a bioinformatic inference from precise active-site mapping plus the DapF-SS literature; it has not been experimentally verified for the *P. putida* enzyme.) Notably, KT2440 encodes **both** subclasses — a canonical DapF-CC (Q88CF3/PP_5228) and this DapF-SS (Q88GD4/PP_3790).

5. Subcellular Localization

DapF acts in the **cytoplasm**. This is consistent with (i) UniProt SUBCELLULAR LOCATION "Cytoplasm" (HAMAP-Rule MF_00197), (ii) UniProt keyword "Cytoplasm," (iii) the absence of any signal peptide, transmembrane segment, or lipobox in the Q88GD4 sequence, and (iv) the fact that its substrate (LL-DAP) and downstream enzymes (LysA, MurE) are cytoplasmic

components of amino-acid and peptidoglycan-precursor metabolism. The *meso*-DAP product is subsequently either decarboxylated in the cytoplasm ($\rightarrow$ L-lysine) or ligated onto the UDP-MurNAc-tripeptide precursor by MurE in the cytoplasm before the lipid-linked precursor is exported for cell-wall assembly.

6. Pathway Context in *P. putida* KT2440

A UniProt proteome survey (taxid 160488) shows KT2440 encodes the **complete succinylase branch** of the DAP pathway upstream of DapF:

Step	Gene(s) / locus	Enzyme (EC)
Aspartate- β -semialdehyde	<i>asd</i> (PP_1989/PP_1992)	ASADH (1.2.1.11)
$\rightarrow$ HTPA	<i>dapA</i> (PP_2639, PP_1237)	4-hydroxy-THDP synthase (4.3.3.7)
$\rightarrow$ THDP	<i>dapB</i> (PP_4725)	HTPA reductase (1.17.1.8)
$\rightarrow$ N-succinyl-THDP	<i>dapD</i> (PP_1530)	THDP N-succinyltransferase (2.3.1.117)
$\rightarrow$ N-succinyl-LL-DAP	<i>dapC</i> (PP_1588)	N-succinyl-LL-DAP aminotransferase (2.6.1.17)
$\rightarrow$ LL-DAP	<i>dapE</i> (PP_1525)	succinyl-DAP desuccinylase (3.5.1.18)
$\rightarrow$ <i>meso</i> -DAP	<i>dapF</i> (PP_3790 = Q88GD4; PP_5228)	DAP epimerase (5.1.1.7)
$\rightarrow$ L-lysine	<i>lysA</i> (PP_2077, PP_5227)	DAP decarboxylase (4.1.1.20)

Critically, **no *ddh* (meso-diaminopimelate dehydrogenase) gene was found**, so the dehydrogenase shortcut that some bacteria use to make *meso*-DAP directly from THDP is absent. Therefore **DapF-catalyzed epimerization is the only route to *meso*-DAP** in KT2440, feeding both peptidoglycan biosynthesis and lysine production. This obligate position, combined with the essentiality of *meso*-DAP for Gram-negative cell-wall cross-linking, explains why the DAP/lysine pathway is a validated **antibacterial drug target** — the pathway "provides both lysine for protein synthesis and meso-diaminopimelate for construction of the bacterial peptidoglycan cell wall" and is absent in mammals (Hutton et al. 2003,

P 12570844

Girodeau et al. 1986,

P 3086556

Paralogy note (mechanistically distinct paralogs). KT2440 encodes **two divergent *dapF* paralogs** that appear to belong to *different mechanistic subclasses*: the target **PP_3790 (Q88GD4, 268 aa)** is a **putative DapF-SS** (catalytic Ser70/Ser205), while **PP_5228 (Q88CF3, 276 aa)** is a **canonical DapF-CC** (catalytic Cys75/Cys219). They share only ~38–40% pairwise identity (global Needleman-Wunsch) yet carry identical UniProt FUNCTION/PATHWAY/localization annotations. PP_5228 lies adjacent to a *lysA* paralog (*lysA-II*, PP_5227), suggesting a lysine-operon-associated copy, whereas PP_3790 is a stand-alone locus. The physiological division of labor — and whether both are catalytically active in vivo — has not been experimentally resolved; the coexistence of a DapF-CC and a DapF-SS in one genome is itself a noteworthy observation.

7. Supported and Refuted Hypotheses

Supported (by orthologous experimental evidence + Q88GD4 sequence/annotation): - H1 — DapF catalyzes LL-DAP $\rightleftharpoons$ *meso*-DAP (EC 5.1.1.7): **Supported.** - H2 — Cofactor-independent (PLP-independent) two-base mechanism: **Supported.** (Refinement: for Q88GD4 the two-base pair is inferred to be **serines, not cysteines** — a DapF-SS enzyme; see §4.) - H3 — Monomeric two-domain fold with substrate-induced domain closure: **Supported.** - H4 — Narrow DAP substrate specificity (sub-mM K_m): **Supported.** - H5 — Cytoplasmic localization: **Supported.** - H6 — DapF is the obligate *meso*-DAP source in KT2440 (no *ddh*): **Supported (bioinformatic).**

Refuted / ruled out: - DapF is **not** PLP-dependent (distinguishes it from Ala/Glu racemases). - DapF is **not** the *phzF*-type isomerase despite the shared fold (lacks that activity; different catalytic residues).

8. Evidence Quality and Limitations

- **Direct experimental evidence in *P. putida* KT2440 is lacking.** Q88GD4 is annotated PE:3 ("inferred from homology"). All kinetic/mechanistic/structural data come from

well-characterized orthologs (*E. coli*, *H. influenzae*) and family-wide studies; transfer of these properties is justified by high family conservation (HAMAP MF_00197) but has not been verified with the *P. putida* enzyme itself.

- **Substrate-specificity constants** (Km, Ki) are from *E. coli* DapF, not the *P. putida* protein.
- **Catalytic residues / DapF-SS assignment:** the identification of Ser70/Ser205 as the catalytic pair (making Q88GD4 a DapF-SS enzyme) is a **bioinformatic inference** from precise mapping to experimentally annotated orthologs plus the DapF-SS literature. It has not been confirmed by an experimental *P. putida* structure or by mutagenesis, and the enzyme's activity has not been directly assayed.
- **Paralog function:** the physiological roles of PP_3790 (DapF-SS) vs PP_5228 (DapF-CC) — redundancy, differential expression, conditional use, or whether one is catalytically silent — are untested.

Future directions: recombinant expression and steady-state kinetics of PP_3790 vs PP_5228; Ser70Ala/Ser205Ala (and Ser→Cys) mutagenesis to confirm the DapF-SS catalytic pair; single/double knockouts to test DAP auxotrophy, essentiality, and functional redundancy; an experimental or AlphaFold structure to verify active-site geometry.

9. Key References

- Koo & Blanchard 1999, *Biochemistry* — chemical mechanism of *H. influenzae* DAP epimerase ([P 10194362](#)).
- Cirilli et al. 1998, *Biochemistry* — first DapF crystal structure; two domains, catalytic Cys73/Cys217 ([P 9843410](#)).
- Pillai et al. 2007, *J. Mol. Biol.* — mutant structures; open/closed conformational cycle ([P 17889830](#)).
- Lam et al. 1988, *Biochemistry/J. Med. Chem.* — DAP analogs; Km and inhibitor Ki for *E. coli* DapF ([P 3042781](#)).

- Lamer et al. 2025 — DapF inhibition by slow-binding α -methyl amino acids

(P 40299312);

DapF-SS serine-variant epimerases

(P 40774471)

— the subclass to which Q88GD4 is assigned.

- Hutton et al. 2003, *Mini Rev. Med. Chem.* — lysine-pathway enzymes as antibacterial targets

(P 12570844).

- Girodeau et al. 1986 — DAP-pathway inhibitors; DAP epimerase as target

(P 3086556).

- UniProt Q88GD4 (HAMAP-Rule MF_00197); Pfam PF01678; InterPro IPR001653; TIGR00652.